

PRD:Product Requirements Document (PRD)

DSA Visualizer Platform

Version: 1.0

Organization: Taskintern Softworks

Product Category: Educational Technology (EdTech)

Status: Development Ready

Document Owner: Founder, Taskintern Softworks

Intended Audience: Software Engineering Interns

Executive Summary

Taskintern is building an interactive web-based learning platform that enables students to understand Data Structures and Algorithms through visualization, experimentation, and hands-on exploration.

Traditional learning resources often rely on static diagrams and theoretical explanations. While these approaches introduce concepts, they frequently fail to help learners build an intuitive understanding of how data structures operate internally and how algorithms execute step-by-step.

The DSA Visualizer Platform aims to bridge this gap by providing an interactive environment where users can perform operations, observe execution in real time, analyze complexity, and connect theoretical concepts with practical implementations.

This product is intended to become a foundational learning module within the Taskintern learning ecosystem.

Problem Statement

Students struggle to understand:

- Understanding internal behavior of data structures
- Visualizing algorithm execution
- Comparing performance trade-offs between algorithms
- Relating interview questions to actual implementations
- Retaining concepts learned through static resources

Most existing resources rely on static diagrams, making concepts difficult to visualize and retain.

Product Goal

Create an interactive web application that enables users to:

- Learn DSA concepts visually
- Interact with data structures in real time
- Observe algorithm execution step-by-step
- Understand time and space complexity
- Explore implementation logic
- Improve technical interview preparation

Target Users

Primary Users

- Computer Science Students
- Interview Preparation Candidates
- Beginners learning DSA

Secondary Users

- Educators
- Mentors / Technical Trainers

Success Metrics

The product should enable users to:

- Visualize DSA concepts without external resources
- Complete learning sessions independently
- Understand algorithm execution flow
- Compare algorithms and data structures
- Use the platform as an interview preparation aid

Functional Requirements

Core Product Module: Interactive DSA Learning Experience

Scope

The platform shall provide a single integrated learning module that combines data structure visualization, algorithm visualization, and contextual learning support into one cohesive experience.

Supported Data Structures

The module must support visualization for:

- Arrays
- Linked Lists
- Stacks
- Queues
- Trees
- Binary Search Trees
- Heaps
- Graphs

Supported Algorithms

The module must support visualization for:

Sorting Algorithms

- Bubble Sort
- Selection Sort
- Insertion Sort
- Merge Sort
- Quick Sort

Searching Algorithms

- Linear Search
- Binary Search

Graph Algorithms

- Breadth First Search (BFS)
- Depth First Search (DFS)

User Actions and Controls

Users should be able to interact with the module through the following actions:

- Insert
- Delete
- Search
- Traverse
- Reset

For algorithm execution, users should also be able to:

- Play
- Pause
- Step Forward
- Restart
- Adjust Execution Speed

Learning Experience Requirements

Each data structure and algorithm must include the following learning components:

Concept

A simple and clear explanation of the topic.

Working

A step-by-step breakdown of how the structure or algorithm behaves during execution.

Complexity Analysis

Display:

- Best Case
- Average Case
- Worst Case
- Space Complexity

Expected Behavior

The module should behave as a real-time interactive learning environment with the following characteristics:

- Visualizations update immediately after every operation
- Current state is always clearly visible
- Animations illustrate structural changes and algorithm flow
- Active elements are highlighted during execution
- Comparisons, swaps, traversals, and state transitions are visually represented
- Invalid operations display meaningful feedback
- Learning content remains accessible alongside the visualization

Product Outcome

This single module should enable users to learn, experiment, and understand DSA concepts without relying on external resources, while also serving as a practical interview preparation tool.

User Experience Requirements

The platform should feel modern, intuitive, educational, and engaging.

Requirements:

- Responsive Design
- Smooth Animations
- Fast Interaction
- Clear Typography
- Accessible Navigation

Nice to Have:

- Keyboard Shortcuts
- Full-Screen Visualization Mode
- Theme Customization
- Interactive Tutorials

Technical Requirements

Frontend:

- Next.js
- TypeScript
- Tailwind CSS
- Use any component library if needed.

Visualization:

- SVG or Canvas-based rendering

Deployment:

Developer Choice

Open Innovation Section

This section is intentionally undefined.

Contributors are encouraged to think beyond the requirements and propose features that improve learning outcomes.

Potential examples:

- AI-generated explanations
- Interactive quizzes
- Learning paths
- Progress tracking
- Custom input playground
- Algorithm comparison mode
- Shareable visualizations
- Code execution visualization
- Voice explanations

Additional ideas are welcome.

Deliverables

Required:

- Source Code
- Documentation
- Live Deployment
- Setup Instructions

Optional:

- Product Walkthrough Video
- Feature Proposal Document
- Architecture Diagram

Ownership & Contribution

This project is part of an active product initiative within Taskintern.

Each intern is expected to independently design, develop, and deliver their implementation based on the requirements outlined in this document, while applying strong technical judgment and product thinking throughout the process.

Interns should take full ownership of their work, proactively identify opportunities for improvement, and make thoughtful decisions that strengthen the learning experience and overall product quality.

While this document defines the minimum product requirements, contributors are encouraged to propose and implement additional features that align with the product vision and improve user outcomes.